Formulario de Mecánica Celeste (Temas 1 y 2)

Tema 1

Fuerza y ecuación de movimiento

$$F(x) = -\frac{GMm}{|x|^2}\frac{x}{|x|}, \quad x \in \mathbb{R}^3 \setminus \{0\}$$

$$\mu = GM, \qquad f(r) = -\frac{\mu}{r^2}$$

$F(x) = f(|x|)\dfrac{x}{|x|}$ (campo de fuerzas), $\qquad f : (0,+\infty) \to \mathbb{R}$

$$\ddot{x}(t) = f\big(|x(t)|\big)\frac{x(t)}{|x(t)|} \qquad \ddot{x}(t) = -\frac{\mu x}{|x(t)|^3} \quad \text{(newtoniano)}$$

Energía

$$E_c = \tfrac{1}{2}mv^2, \qquad E_{pg} = mgh$$

$$F = -\nabla V \ \text{ en } \ \Omega \subseteq \mathbb{R}^3 \text{ abierto}$$

$$E = \frac{|\dot{x}|^2}{2} + V(x) = \frac{|\dot{x}|^2}{2} - \Phi(|x|), \quad \Phi(r) = \int_1^r f(s)\,ds$$

$$E = \frac{|\dot{x}|^2}{2} - \frac{\mu}{|x|} \quad \text{(caso newtoniano)}$$

Momento angular y área

$$c(t) = x(t) \wedge \dot{x}(t) = r^2\dot{\theta}\,(0,0,1)$$

$$x = r(\cos\theta, \operatorname{sen}\theta) \ \Rightarrow \ \dot{\theta} = x_1\dot{x}_2 - \dot{x}_1x_2$$

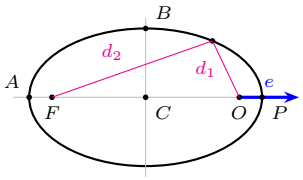
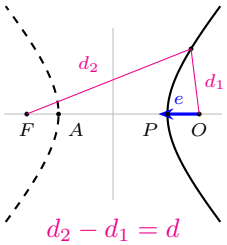
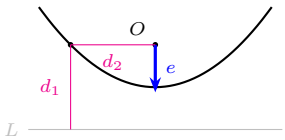
$$\text{Área}(D) = \frac{1}{2}\int_{t_0}^{t_1} r^2\dot{\theta}\,dt = \frac{1}{2}|c|\,(t_1 - t_0)$$

TEOREMA DE PROLONGACIÓN ($c = 0$)

i) $\lim\limits_{t\to\omega} r(t)^2 + \dot{r}(t)^2 = +\infty$ $\qquad$ ii) $\exists\, t_n \to \omega : \ r(t_n) \to 0$

Tema 2

$$\boxed{|x| + \langle x, e\rangle = k} \qquad e = \frac{1}{\mu}\dot{x} \wedge c - \frac{x}{|x|} \qquad k = \frac{|c|^2}{\mu}$$

Elipse ($\lvert e\rvert < 1$)	**Hipérbola** ($\lvert e\rvert > 1$)	**Parábola** ($\lvert e\rvert = 1$)
 $d_1 + d_2 = d$ $\quad$ Área $= ab\pi$	 $d_2 - d_1 = d$	 $d_1 = d_2 = d$ $\quad L : \langle x, v\rangle + d = 0$
$e = -\dfrac{1}{d}F$	$e = \dfrac{1}{d}A$	$e = -v$
$k = \dfrac{c^2 - \lvert F\rvert^2}{2d}$	$k = \dfrac{\lvert A\rvert^2 - d^2}{2d}$	$k = d$
$\lvert BC\rvert = \sqrt{1-\varepsilon^2}\,a, \quad \lvert OC\rvert = \varepsilon a$ $\lvert OP\rvert = \dfrac{k}{1+\varepsilon}, \quad \lvert OA\rvert = \dfrac{k}{1-\varepsilon}$ $a = \dfrac{k}{1-\varepsilon^2}, \quad b = \sqrt{1-\varepsilon^2}a$	$\varepsilon = \dfrac{\lvert FO\rvert}{\lvert AP\rvert} \quad \lvert OP\rvert = \dfrac{k}{1+\varepsilon}$	—

Parametrización:

$$\theta \mapsto \frac{k}{1+\varepsilon\cos(\theta-\omega)}(\cos\theta, \operatorname{sen}\theta), \qquad \gamma : \mathbb{R} \to \mathbb{R}^2, \quad \gamma : (\omega-\pi, \omega+\pi) \to \mathbb{R}^2, \quad \gamma : (\omega-\ell, \omega+\ell) \to \mathbb{R}^2, \quad \ell \in (\tfrac{\pi}{2}, \pi)$$

$$P = \frac{2\pi}{\sqrt{\mu}}\,a^{3/2}, \qquad \mu^2(\varepsilon^2 - 1) = 2h|c|^2$$

$\boxed{i, \Omega, a, \varepsilon, \omega, T}$ solución elíptica

$$\begin{cases} x_1 = a\cos u - \varepsilon \\ x_2 = a\sqrt{1-\varepsilon^2}\,\operatorname{sen} u \\ u - \varepsilon\operatorname{sen} u = \dfrac{2\pi}{P}(t - t_0) \end{cases}$$

ECUACIÓN DE KEPLER

$$P = t(2\pi) - t(0)$$

$$f_\varepsilon(u) = u - \varepsilon\operatorname{sen} u = \xi \ \Rightarrow \ u_{n+1} = u_n - \frac{f_\varepsilon(u_n) - \xi}{f_\varepsilon'(u_n)}$$

$$J_n(x) = \frac{1}{\pi}\int_0^\pi \cos(nu - x\operatorname{sen} u)\,du \quad \text{(Bessel, índice } n)$$

$$u(t) = \underbrace{M + \sum_{n=1}^{\infty}\frac{2}{n}J_n(n\varepsilon)\operatorname{sen}(nM)}_{K_\varepsilon(\xi)}, \qquad M = \tfrac{2\pi}{P}(t - t_0)$$

Órbitas no newtonianas

$$\begin{cases} f(r) = \ddot{r} - \dfrac{|c|^2}{r^3} \\ \dot{\theta} = \dfrac{|c|}{r^2}, \ r > 0 \end{cases} \Rightarrow \quad \frac{d^2p}{d\theta^2} + p = -\frac{f(1/p)}{p^2|c|^2}$$

con $p(\theta) = \dfrac{1}{r(t(\theta))}$